

HARSH AGALE

AI/ML ENGINEER

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TECHNICAL SKILLS

Languages

Python JavaScript Java C SQL

AI / ML

PyTorch TensorFlow LangChain LangGraph LCEL Hugging Face PEFT LoRA/QLoRA RAG FastAPI XGBoost LLM Integration NLP sentence-transformers

LLM Ops

MLflow W&B DVC RAGAS Langfuse LangSmith Evidently CrewAI DPO/TRL

Vector DBs

Qdrant ChromaDB

Data & Backend

NumPy Pandas Streamlit REST APIs OAuth 2.0

DevOps & Cloud

Docker AWS CI/CD Kubernetes Git

Frontend

React + Vite Angular

CERTIFICATIONS

AWS Cloud Practitioner
Data Science Professional
Deep Learning
Machine Learning

OPEN SOURCE

Merged PR — LangChain / PEFT / RAGAS (good first issue)

PUBLISHED ARTICLES

- Fine-tuning LLMs on a budget: QLoRA explained
- LangGraph → 3-agent system: a migration story
- Multimodal RAG: making chatbots read charts

PROFESSIONAL SUMMARY

B.Sc. Computer Science graduate and AI Engineer with hands-on expertise in LLM fine-tuning (QLoRA/LoRA), multi-agent orchestration, RAG evaluation, and production MLOps pipelines. Proficient in PyTorch, Hugging Face, LangChain, LangGraph, CrewAI, Docker, and AWS. Built and deployed fine-tuned LLMs on Hugging Face Hub, RAGAS-evaluated RAG systems with Langfuse observability, multi-agent self-healing systems with W&B tracing, and multimodal RAG pipelines. Open source contributor. 100+ DSA problems on LeetCode.

WORK EXPERIENCE

AI Engineer Intern

Dec 2025 – Mar 2026

Glaxm Infotech

- Engineered and deployed production-grade AI/ML solutions contributing to critical real-world systems
- Implemented LLM integration and end-to-end data pipelines using Python, LangChain, and FastAPI
- Collaborated with cross-functional teams to drive full SDLC for AI-powered features

FEATURED PROJECTS

RAG Chatbot — Upgraded with Qdrant, Fine-tuned Embeddings & RAGAS Evaluation

Python FastAPI Qdrant sentence-transformers LangChain RAGAS Langfuse React + Vite

- Migrated from ChromaDB to Qdrant with hybrid dense+sparse (BM25) retrieval; fine-tuned domain embedding model using SBERT/MultipleNegativesRankingLoss, improving retrieval quality measurably
- Integrated RAGAS evaluation pipeline (faithfulness, answer relevancy, context precision/recall) and Langfuse observability — wired into FastAPI via /evaluate and /evaluate/status endpoints

Fine-tuned LLM — QLoRA + DPO Alignment

PyTorch Hugging Face PEFT QLoRA TRL DPO MLflow W&B FastAPI Docker

- Fine-tuned Llama-3 8B on domain Q&A dataset using QLoRA (4-bit quantization, bitsandbytes) with PEFT; tracked all experiments with MLflow and W&B; published adapter weights to Hugging Face Hub
- Applied DPO alignment with TRL library on top of SFT model; evaluated base vs fine-tuned vs DPO outputs using BLEU, ROUGE, and LLM-as-judge; deployed as Dockerized FastAPI inference server

Multi-Agent Self-Healing System

Python CrewAI LangGraph Qdrant W&B FastAPI

- Refactored LangGraph self-healing pipeline into 3 specialised CrewAI agents: Log Parser, Root Cause Analyst, and Patch Generator — each with dedicated tools, memory, and Qdrant RAG context
- Added W&B tracing to log every agent decision and tool call; each agent handoff tracked as a span in the W&B dashboard for full observability

Multimodal RAG System

Python ColPali Qdrant FastAPI LangChain GPT-4o Vision

- Extended RAG pipeline to handle PDFs with embedded charts, tables, and images using ColPali for page-level image embedding alongside text chunks in a unified Qdrant collection
- Tested on financial reports and research papers — demonstrated accurate retrieval of chart-heavy pages that pure text RAG missed entirely

AI-Powered System Log & Incident Investigator

Python AWS Docker Angular LangChain

- Engineered scalable log analysis system processing millions of lines, reducing investigation time by 70%
- Deployed on AWS with Dockerized microservices; reduced inference costs by 40% through pre-filtering

Autonomous Calendar Agent & Customer Churn Prediction

LangChain OAuth 2.0 XGBoost Scikit-learn RealTimeSTT

- Voice-enabled AI agent for Google Calendar scheduling via RealTimeSTT + LangChain + OAuth 2.0
- Churn prediction model using XGBoost & Logistic Regression; achieved 85%+ accuracy with SMOTE and GridSearchCV

Duplicate Detection / Sentiment Analysis / Student Performance

TF-IDF LSTM Random Forest Python Scikit-learn

- TF-IDF + cosine similarity deduplication pipeline; LSTM sentiment classifier (positive/negative/neutral); Random Forest grade predictor

EDUCATION

B.Sc. Computer Science

2023 – 2026

Nowrosjee Wadia College, Pune